Introduction

Helyx is a lightweight, high-performance replication solution designed for **real-time heterogeneous database replication**. It provides a simple CLI-driven experience and abstracts the complexity of setting up connectors, schema registry, and replication pipelines.

Key highlights of **Helyx**:

- **Heterogeneous Replication**: Supports replication across multiple database systems (e.g., PostgreSQL ↔ PostgreSQL, Oracle ↔ Oracle, Oracle → PostgreSQL, Oracle → MySQL, Oracle → MongoDB, Oracle → Snowflake).
- **High Throughput**: Capable of handling **hundreds of thousands of transactions per second** across distributed data centers.
- **Schema Evolution Handling**: Automatically adapts to schema changes like column additions and keeps source and target databases in sync.
- **Zero-Downtime Replication**: Supports **ad-hoc snapshots** and live streaming without interrupting production workloads.
- **Monitoring & Control**: Built-in CLI commands allow users to check replication lag, list connectors, view tasks, and monitor end-to-end replication health.
- **Deployment Friendly**: Delivered as **self-contained executables and RPM packages**, making installation and upgrades straightforward.
- **Version-Aware Replication**
  Helyx supports replication between **different versions of the same technology stack** (e.g., PostgreSQL 11 → PostgreSQL 15 or EDB 12 → EDB 14 or Oracle11g → Oracle19c). This ensures smooth migrations and upgrades without downtime.

With Helyx, enterprises can achieve **enterprise-grade replication** with lower operational overhead, faster setup, and proven reliability compared to traditional tools.

Helyx Installation Guide

1. Overview

Helyx is distributed as an **RPM package** to simplify installation and management. By default, Helyx is installed in the directory **/var/lib/helyx.**

This section describes the steps to install Helyx from a standard YUM/DNF repository as well as from a local repository if you download the RPM manually.

2. Prerequisites

Before beginning installation, ensure the following prerequisites are met:

- **Operating System**: Linux (RHEL, CentOS, Rocky Linux, Oracle Linux, Fedora, etc.).
- **Privileges**: Root or sudo access.
- **Package Manager**: yum or dnf depending on your OS version.
- **Network Access** (for online installation): Access to the configured YUM repository

Installing Helyx from a Local Repository

Create Local Repository Directory:

mkdir -p /opt/localrepo/helix

cp helyx-1.1.0.rpm /opt/localrepo/helyx/

Create Repository Metadata:

cd /opt/localrepo/helix

createrepo .

Add Local Repository Entry:

Create **/etc/yum.repos.d/helyx-local.repo** with the following:

[helyx-local]

name=Helyx Local Repository

baseurl=file:///opt/localrepo/helyx

enabled=1

gpgcheck=0

Install from Local Repository

sudo yum install helyx -y

Post installation verifies and ensures binaries are installed under **/var/lib/helyx** location.

PostgreSQL Replication Setup

Target: Set up real-time replication between PostgreSQL/EDB source and destination with different versions as well as same version.

Prerequisites

- Java 11 or above
- Replication Binaries
- PostgreSQL or EDB installed at source and destination
- JDBC connectors come pre-build with Binaries
- Config tools: `tabletorepl`, `serverconfig and configservice`
- **wal_level = logical** for both Source and Destination postgresql/EDB Databases

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Directory Structure

```
/var/lib/helyx/replconfig
    ├── publication.config
    ├── subscription.config
    ├── tablelist.config
    ├── tabletorep
    └── serverconfig and configservice
```

Step 1: Setup Replication & Related Services

Ensure these services are configured and started:

1. **Sync-manager service**:

   ```
   ./configservice -start sync-manager
   ```

2. **Broker Service**:

   Edit /var/lib/helyx/ server.properties file. Change the parameters below:

listeners = PLAINTEXT://your.host.name:9092

your.host.name will be the IP address of the server where helyx is running.

Now start broker service:

```
./configservice -start broker
```

3. **Registry Service**:

   Edit /var/lib/helyx/plugins/confluent-7.5.0/etc/schema-registry/schema-registry.properties and ensure:

   - kafkastore.bootstrap.servers=PLAINTEXT:// <your.host.name>:9092
   - listeners=http://0.0.0.0:8081

   your.host.name will be the IP address of the server where helyx is running.

```
./configservice -start schemaregistry
```

4. **Connect Service**:

   Edit /var/lib/helyx/config/connect-distributed.properties and set:

```
bootstrap.servers=<your.server.ip>:9092
key.converter.schema.registry.url=http://<your.server.ip>:8081
value.converter.schema.registry.url=http://<your.server.ip>:8081
```

```
./connectservice -start connect
```

Step 2: Configure Source Database

Creating Helyx User in Source Database:

Overview

In order for Helyx to perform replication, a dedicated database user with appropriate privileges must be created. This section describes the SQL commands required to create the **helyx** role, schema, and associated privileges on the source database

1. **Create Replication Role**

```
CREATE ROLE helyx WITH LOGIN REPLICATION PASSWORD 'password';
```

2. **Create Schema**

```
CREATE SCHEMA IF NOT EXISTS helyx;
```

3. **Grant Database Access**

```
GRANT CONNECT ON DATABASE edb TO helyx;

GRANT CREATE ON DATABASE edb TO helyx;
```

4. **Grant Table-Level Privileges**

```
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA helyx TO
helyx;

ALTER DEFAULT PRIVILEGES IN SCHEMA helyx

GRANT SELECT, INSERT, UPDATE, DELETE ON TABLES TO helyx;
```

Change publication.config file:

Edit `/var/lib/helyx/replconfig/publication.config` with the following keys:

```
"name": "",                          // Unique identifier
"tasks.max": "1",                   // Parallelism level
"database.type": "edb",             // edb or postgresql
"database.hostname": "",            // Source DB IP
"database.port": "5444",            // Source DB Port
"database.password": "",            // Source DB “helyx” user password
"database.dbname": "edb",           // Database name
"bootstrap.ip": "",                 // Broker Server IP
"include.schema.list": "schema1,schema2,schema3",
"schema.history.internal.kafka.bootstrap.servers": "<serverIP>:9092",
"publication.name": "",             // Logical replication publication name
"snapshot.mode": "initial",
"database.history.kafka.bootstrap.servers": "<serverIP>:9092",
"database.history.kafka.topic": "dbhistory.<Topicname>",
"topic.prefix": "",                 // Prefix for Replication topics
"max.batch.size": "2048",
"max.queue.size": 50000,
"poll.interval.ms": "1000",
"producer.batch.size": "32768",
"producer.max.request.size": "10485760"
```

Grant Source Permissions Automatically

Run this to apply necessary grants at source:

```
Enter comma-separated Table list at
/var/lib/helyx/replconfig/tablelist.config.
These are the tables which you want to replicate.
Example: Schema1.Table1,
         Schema2.Table2,
         Schema2.Table3
Then Run tabletorepl.jar for Necessary Privileges:
./tabletorepl.jar -s <path_to_publication.config> -tables <path_to_tablelist.config>
```

For command help Run

```
./tabletorepl.jar -help
```

Step 3: Configure Destination Database

Creating Helyx User in Destiantion Database:

Overview

In order for Helyx to perform replication, a dedicated database user with appropriate privileges must be created. This section describes the SQL commands required to create the **helyx** role, schema, and associated privileges on the **destination** database.

1. **Create Replication Role**

   ```
   CREATE ROLE helyx WITH LOGIN REPLICATION PASSWORD 'password';
   ALTER ROLE helyx WITH REPLICATION;
   ```

2. **Create Schema**

   ```
   CREATE SCHEMA IF NOT EXISTS helyx AUTHORIZATION helyx;
   ```

3. **Grant Database Access**

   ```
   GRANT CONNECT ON DATABASE edb TO helyx;
   GRANT CREATE ON DATABASE edb TO helyx;
   ```

Granting Schema Privileges for Helyx User

For Helyx to replicate tables from specific schemas, the replication user (helyx) must have appropriate privileges on those schemas.
The schemas that require privileges at destination database are defined in the `include.schema.list` parameter of the `publication.config` file.

1. **Grant USAGE on Each Schema**

```
GRANT USAGE ON SCHEMA <schema_name> TO helyx;
```

2. **Grant Table-Level Privileges**

```
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA
<schema_name> TO helyx;

GRANT CREATE ON SCHEMA <schema_name> TO helyx;

ALTER SCHEMA <schema_name> OWNER TO helyx;

ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT ALL ON TABLES TO
helyx;

ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT SELECT, INSERT,
UPDATE, DELETE ON TABLES TO helyx;
```

Note: Please note that username will be same as source and destination database with same password.

Change subscription.config file:

Edit `/var/lib/helyx/replconfig/subscription.config` with the following keys:

```
"name": "",   // Unique identifier name
"tasks.max": "1",
"connection.url": "jdbc:postgresql://<Dest_ServerIP>:5444/edb?user=helyx&pass
word=<password>",
"bootstrap.ip": "",  // Kafka IP
"retry.backoff.ms": "1000",
"consumer.max.poll.records": "1000",
"consumer.fetch.max.bytes": "10485760",
"batch.size": "5000",
"max.retries": "10"
```

Note: User name **helyx** will **NOT** be changed

Step 4: Use CLI Tool for Replication Management

Here are common usages of **`serverconfig`**:

Action	Command Format
Create publication	`serverconfig -createpub -s publication.config -tables tablelist.config`
Create subscription	`serverconfig -createsub -s subscription.config -pub publication.config`
Add tables to publication	`serverconfig -addtabletopub -s publication.config -tables tablelist.config`
Trigger ad-hoc snapshot	`serverconfig -adhocsnapshot -s publication.config -tables table1,table2`
Monitor publication status	`serverconfig -status -pub -s publication.config`
Monitor subscription status	`serverconfig -status -sub -s subscription.config`
Remove publication	`serverconfig -remove -pub -s publication.config`
Remove subscription	`serverconfig -remove -sub -s subscription.config`
Remove table from replication	`serverconfig -removetablefromrepl -s publication.config -tables table1,table2`
List connectors	`serverconfig -listconnectors -s publication.config`
List tasks	`serverconfig -listtask -s publication.config`
Lag statistics	`serverconfig -lagstat -s subscription.config`
List replicated tables	`serverconfig -tablelist -s publication.config`

🎯 Step 5: Set Up Destination Database (Manual Grants)

On **destination DB**, execute:

```
CREATE ROLE helyx WITH LOGIN PASSWORD '<password>';
ALTER ROLE helyx WITH REPLICATION;
GRANT CREATE ON DATABASE <Database Name> TO helyx;
CREATE SCHEMA IF NOT EXISTS helyx AUTHORIZATION helyx;

GRANT USAGE ON SCHEMA <schema_name> TO helyx;
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA <schema_name> TO
helyx;
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA <schema_name> TO
helyx;
GRANT CREATE ON SCHEMA <schema_name> TO helyx;
GRANT CREATE ON SCHEMA <schema_name> TO helyx;
ALTER SCHEMA <schema_name> OWNER TO helyx;
```

```
ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT ALL ON TABLES TO helyx
;
ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT ALL ON TABLES TO helyx
;
ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT USAGE ON SEQUENCES TO
helyx;
ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT USAGE ON SEQUENCES TO
helyx;
ALTER DEFAULT PRIVILEGES IN SCHEMA <schema_name> GRANT SELECT, INSERT, UPDATE
, DELETE ON TABLES TO helyx;
```

Step 6: Use CLI Tool for Starting Replication between servers:

Action	Description
Create publication	Creates a logical publication on the source DB using `publication.config` and tables from `tablelist.config`.
Create subscription	Sets up the sink connector using `subscription.config`, and links it to the Kafka topics defined in the publication.
Add tables to publication	Adds new tables to the existing source-side publication so that changes in those tables start replicating.
Trigger ad-hoc snapshot	Triggers a one-time snapshot for specified tables, even after initial replication setup. Useful for re-syncing.
Monitor publication status	Shows current status of the publication connector (active, paused, failed, etc.).
Monitor subscription status	Displays the running status of the Subscription and its tasks.
Remove publication	Completely Stops removes the publication and associated configuration.
Remove subscription	Stops and deletes the Subscription configuration.
Remove table from replication	Removes selected tables from the ongoing replication setup.
List connectors	Lists all running Kafka connectors on the connect cluster. Helps in validation and monitoring.

Action	Description
List tasks	Displays tasks of a connector, their IDs, and current status (running, failed, paused).
Lag statistics	Shows replication lag metrics to monitor delay in syncing data to the destination.
List replicated tables	Lists all tables currently under replication from the source config.

📌 Command Reference:

```
serverconfig -createpub -s publication.config -tables tablelist.config
serverconfig -createsub -s subscription.config -pub publication.config
serverconfig -addtabletopub -s publication.config -tables tablelist.config
serverconfig -adhocsnapshot -s publication.config -tables table1,table2
 serverconfig -status -pub -s publication.config
 serverconfig -status -sub -s subscription.config
 serverconfig -remove -pub -s publication.config
 serverconfig -remove -sub -s subscription.config
 serverconfig -removetablefromrepl -s publication.config -tables table1,table
2
 serverconfig -encryptPassword -s publication.config
 serverconfig -listconnectors -s publication.config
 serverconfig -listtask -s publication.config
 serverconfig -lagstat -s subscription.config
 serverconfig -tablelist -s publication.config
```

✅ Once *configuration* and *grants* are ready, start the **replication**:

▶️ Start the Publication

```
serverconfig -createpub -s publication.config -tables tablelist.config
```

▶️ Start the Subscription

```
serverconfig -createsub -s subscription.config -pub publication.config
```

▶️ You can monitor the connector statuses:

```
serverconfig -status -pub -s publication.config
serverconfig -status -sub -s subscription.config
```

🚀 Final Checklist

Item	Status
sync-manager, broker, Connect, Schema Registry running	✅

Item	Status
Source `publication.config` configured	✅
Destination `subscription.config` configured	✅
Grants applied to source and destination DBs	✅
Topics appearing in Kafka	✅
Messages flowing to sink	✅

🔄 Schema Evolution: Add Column to Replicated Table

If you want to add a new column to a table that is already being replicated, follow these steps carefully:

🧬 Purpose:

To safely add a **new column** to an existing replicated table and synchronize the schema with downstream systems.

☑️ Steps:

3. **Stop the Connect Worker service**:

```
systemctl stop connect.service
```

   This ensures that no replication occurs during the schema update.

4. **Add the column in the source database**: Connect to the *source* database and run:

```
ALTER TABLE <schema>.<table_name> ADD COLUMN <new_column_name> <datatype>;
```

5. **Start the Connect Worker service**:

```
systemctl start connect.service
```

   The connector will reload the schema change and start applying changes again.

6. **Trigger an ad-hoc snapshot for the updated table**: This will send the entire current state of the table (with the new column) to the existing topic.

```
serverconfig -adhocsnapshot -s publication.config -tables <table_name>
```

This ensures the new column is picked up and replicated to the destination database.

🧪 Optional: Monitoring & Troubleshooting

- Check Kafka logs:

```
tail -f /var/lib/helyx/logs/connect.log
```

- Validate schema Registry at: `http://<schema-registry>:8081/subjects`
- Monitor replication lag:

```
serverconfig -lagstat -s subscription.config
```

📞 Need Help?

Run:

```
serverconfig -help
```

For all supported actions.

> **Note**: Always test on a staging environment before applying to production.

Happy Replicating! 🚀